(iii) No solution (iv) $x=\frac{1}{2}(s-t+2), y=s, z=t$ and $s,t\in\mathbb{R}$

2. (i) $k=1$ (ii) $k\neq 1, k\neq -2$ (iii) $k=-2$

3. (i) $\lambda = 5$ and $\mu\neq 9$ (ii) $\lambda\neq 5$ and $\mu\in\mathbb{R}$ (iii) $\lambda=5, \mu=9$

Exercise 1.7

1. (i) $x=-t, y=-2t, z=t, t\in\mathbb{R}$ (ii) Trivial solutions only

2. (i) $\lambda\neq 8$ (ii) $\lambda=8$

3. $2C_2H_6+7O_2\rightarrow 6H_2O+4CO_2$

Exercise 1.8

1	2	3	4	5	6	7	8	9	10
(2)	(3)	(2)	(3)	(4)	(2)	(4)	(4)	(2)	(1)
11	12	13	14	15	16	17	18	19	20
(2)	(4)	(1)	(2)	(4)	(3)	(2)	(1)	(4)	(4)
21	22	23	24	25					
(2)	(4)	(4)	(4)	(1)					

Exercise 2.1

1. $-1-i$ 2. $1+i$ 3. 0 4. 0

5. 1 6. $1-i$

Exercise 2.2

1. (i) $4+i$ (ii) $8-i$ (iii) $7+5i$ (iv) $1+17i$

(v) $15+8i$ (vi) $15+8i$

3. $x=-1,\ y=1$

Exercise 2.3

3. $-z_1=-2-5i$, $z_1^{-1}=\frac{1}{29}(2-5i)$

$-z_2=3+4i$, $z_2^{-1}=\frac{1}{25}(-3+4i)$

$-z_3=-1-i$, $z_3^{-1}=\frac{1}{2}(1-i)$

Exercise 2.4

1. (i) $7-5i$ (ii) $\frac{5}{4}(1-i)$ (iii) $\frac{2}{5}-\frac{14i}{5}$

2. (i) $\frac{x}{x^2+y^2}$ (ii) y (iii) $-y-4$

3. $\frac{1}{25}(-1-2i), \frac{1}{5}(-11+2i)$ (4) $\frac{1}{2}(7-i)$

6. (i) 6 (ii) 3

12th_Maths_EM_Vol1_Answers.indd 283 16-12-2022 12:10:08